JAL SANKET: Literature Review & Architectural Blueprint

Cryospheric Remote Sensing & Early Warning Decision Support for Himalayan GLOFs
Software-Only Research & Development Deliverable | Version 1.0

1. Executive Summary

Glacial Lake Outburst Floods (GLOFs) present an existential threat to downstream infrastructure across the Himalayas. Jal Sanket is designed as a software-only, automated decision support system combining European Space Agency Sentinel-1 Synthetic Aperture Radar (SAR) with multi-criteria hazard evaluation. Crucially, the system decouples raw lake-area expansion from immediate warning generation, incorporating parent glacier slope, moraine stability proxies, and downstream exposure.

2. Critical Assessment of the 15% and 30% Expansion Thresholds

Common assumptions that a 15% or 30% lake expansion signals immediate catastrophic failure are scientifically flawed without contextual geomorphological validation. Natural seasonal ablation causes 8-22% expansion annually. The 2023 South Lhonak disaster demonstrated that sudden avalanche-induced impulse waves cause breaches rather than static hydrostatic expansion alone. Jal Sanket caps area expansion weight at 20% in its composite index.

3. Summary of Key Reviewed Papers

Author / Year	Title	Sensor	Key Finding
Nie et al. (2017)	Glacial lake changes and identification of potentially dangerous glacial lakes in the central Himalayas	Landsat TM/ETM+/OLI	Identified 49 potentially dangerous glacial lakes; lake area expanded 14.1% over 3 decades
Veh et al. (2018)	Determinants of hazard and risk from glacial lake outburst floods in the Himalayas	SRTM DEM, Multi-spectral	GLOF susceptibility driven by moraine geometry, ice-dam degradation, and peak discharge
Allen et al. (2019)	Potentially dangerous glacial lakes across the Tibetan Plateau	Sentinel-2, Landsat	Multi-criteria analytical hierarchy process (AHP) for hazard classification
Strozzi et al. (2012)	Glacier surface velocity retrieval using Sentinel-1 SAR interferometry	Sentinel-1 SAR C-band	SAR C-band penetrates cloud and operates during polar/monsoon night; backscatter drops over calm water
Sattar et al. (2021)	Future GLOF hazard of the South Lhonak Lake, Sikkim Himalaya	High-res DEM, Sentinel-1/2, HEC-RAS	Simulated cascading moraine breach; predicted 2023 disaster parameters accurately

4. Scientific Safety & Mandatory Protocol

Jal Sanket operates under strict safety protocols: all hazard indices are labeled PROVISIONAL. Satellite revisit latencies (6-12 days) and wet-snow speckle require expert verification before public advisory dispatch.